Newton's Laws

ANSWER KEY

Section 1: Newton's Second Law ($F = ma$)

1 A constant resultant force of 36 N acts on a mass of 4 kg. What is the acceleration produced?

a. $9\ \text{m/s}^2$ ✓

b. $144\ \text{m/s}^2$

c. $4\ \text{m/s}^2$

d. $6\ \text{m/s}^2$

$a = F/m = 36/4 = 9\ \text{m/s}^2$.

2 A car of mass 1200 kg is traveling at 25 m/s when the brakes are applied, bringing it to rest in 50 m. Find the magnitude of the braking force.

a. 7500 N ✓

b. 6000 N

c. 9000 N

d. 15000 N

$v^2 = u^2 + 2as \Rightarrow 0 = 25^2 + 2a(50) \Rightarrow 100a = -625 \Rightarrow a = -6.25\ \text{m/s}^2$. Force magnitude $F = ma = 1200(6.25) = 7500\ \text{N}$.

3 An elevator of mass 800 kg accelerates upwards at $1.5\ \text{m/s}^2$. Taking $g = 10\ \text{m/s}^2$, find the tension in the cable.

a. 9200 N ✓

b. 8000 N

c. 6800 N

d. 1200 N

Equation of motion: $T - mg = ma \Rightarrow T = m(g + a) = 800(10 + 1.5) = 800(11.5) = 9200\ \text{N}$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Section 2: Connected Particles over a Pulley

4 Two masses of 3 kg and 2 kg are connected by a light inextensible string passing over a smooth fixed pulley (Atwood machine). Taking $g = 10$ m/s^2, find the acceleration of the system.

a. 2 m/s^2 ✓

b. 1 m/s^2

c. 4 m/s^2

d. 5 m/s^2

Equations: $3g - T = 3a$ and $T - 2g = 2a$. Adding: $(3-2)g = (3+2)a \Rightarrow 10 = 5a \Rightarrow a = 2$ m/s^2.

5 For the system in the previous question, find the tension T in the string.

a. 24 N ✓

b. 25 N

c. 20 N

d. 30 N

$T = 2(g + a) = 2(10 + 2) = 2(12) = 24$ N.

6 What is the total force exerted by the string on the pulley in the previous system?

a. 48 N **downwards** ✓

b. 24 N downwards

c. 50 N downwards

d. 40 N downwards

The string pulls downward on both sides of the pulley with tension $T = 24$ N. Total force on pulley $= 2T = 48$ N downwards.

Section 3: Motion on Rough Inclined Planes

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 A block of mass 5 kg is pulled up a smooth slope inclined at 30° to the horizontal by a force of 40 N acting parallel to the slope. Taking $g = 10$ m/s^2, find the acceleration of the block.

a. 3 m/s^2 ✓

b. 5 m/s^2

c. 2 m/s^2

d. 4 m/s^2

$F_{\text{net}} = P - mg\sin(30^\circ) = 40 - 5(10)(0.5) = 40 - 25 = 15$ N. Then $a = 15/5 = 3$ m/s^2.

8 If the slope in the previous question is rough with $\mu = 0.2$, what is the new acceleration of the block?

a. 1.27 m/s^2 ✓

b. 3.00 m/s^2

c. 2.15 m/s^2

d. 0.85 m/s^2

$R = mg\cos(30^\circ) = 50(\sqrt{3}/2) \approx 43.30$ N. Friction $F_f = \mu R = 0.2(43.30) \approx 8.66$ N. Net force $= 40 - 25 - 8.66 = 6.34$ N. Acceleration $a = 6.34/5 \approx 1.27$ m/s^2.

9 A sled of mass 20 kg slides down an incline of 25° at constant speed. What is the coefficient of kinetic friction μ?

a. $\mu = \tan(25^\circ) \approx 0.466$ ✓

b. 0.423

c. 0.906

d. 0.500

At constant speed, acceleration is zero, so $mg\sin\alpha = \mu mg\cos\alpha \Rightarrow \mu = \tan\alpha = \tan(25^\circ) \approx 0.466$.

Section 4: Connected Systems with Pulleys & Tables

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 A block A of mass 4 kg rests on a smooth horizontal table and is connected by a light string over a smooth pulley at the edge of the table to a hanging block B of mass 1 kg. Taking $g = 10$ m/s^2, find the acceleration.

a. 2 m/s^2 ✓

b. 2.5 m/s^2

c. 1.5 m/s^2

d. 10 m/s^2

Net accelerating force is the weight of B: $m_B g = 1(10) = 10$ N. Total mass $= 4 + 1 = 5$ kg. Acceleration $a = 10/5 = 2$ m/s^2.

11 If the table in the previous system has coefficient of friction $\mu = 0.15$, find the acceleration of the system.

a. 0.8 m/s^2 ✓

b. 1.2 m/s^2

c. 1.5 m/s^2

d. 0.5 m/s^2

Friction on A: $F_f = \mu m_A g = 0.15(4)(10) = 6$ N. Net accelerating force $= 10 - 6 = 4$ N. Total mass $= 5$ kg. Acceleration $a = 4/5 = 0.8$ m/s^2.

12 A car of mass 1000 kg tows a trailer of mass 500 kg with acceleration 1.2 m/s^2. Resistance to motion is 300 N on the car and 200 N on the trailer. Find the tension in the towbar.

a. 800 N ✓

b. 600 N

c. 1100 N

d. 1500 N

Consider the trailer alone: $T - R_{\text{trailer}} = m_{\text{trailer}} a \Rightarrow T - 200 = 500(1.2) \Rightarrow T - 200 = 600 \Rightarrow T = 800$ N.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647